

$$\langle f | g \rangle = \int_1^5 f(x)g(x) dx$$

Gram-Schmidt per $\{1, x, x^2\}$

$$l_1 = 1$$

$$\text{norma } \|l_1\|^2 = \int_1^5 1 dx = 4 \Rightarrow \|l_1\| = 2$$

$$\phi_1 = \frac{1}{2}$$

$$l_2 = x - \frac{\langle x, 1 \rangle}{\langle 1, 1 \rangle} \cdot 1$$

$$\langle x, 1 \rangle = \int_1^5 x dx = \frac{25-1}{2} = 12$$

$$l_2 = x - 3$$

$$\text{norma } \|l_2\|^2 = \int_1^5 (x-3)^2 dx \approx 5,33$$

$$\phi_2 = \frac{x-3}{\sqrt{5,33}}$$

- $l_3 = x^2$ proiezioni su l_1, l_2

proiezione su $l_1 = 1$

$$\langle x^2, 1 \rangle = \int_1^5 x^2 dx = \frac{125-1}{3} = \frac{124}{3}$$

$$\langle 1, 1 \rangle = 4$$

$$\text{proj}_{l_1}(x^2) = \frac{\frac{124}{3}}{4} = \frac{31}{3}$$

proiezione su $l_2 = x-3$

$$\langle x^2, x-3 \rangle = \int_1^5 (x^3 - 3x^2) dx = \left[\frac{x^4}{4} - x^3 \right]_1^5 = \left(\frac{625}{4} - 125 \right) - \left(\frac{1}{4} - 1 \right) =$$

$$= \frac{125}{4} + \frac{3}{4} = 32$$

$$\text{norma } \langle l_2, l_2 \rangle = \int_1^5 (x-3)^2 dx = \frac{16}{3}$$

$$\text{projekcia } l_2(x^2) = \frac{\frac{32}{16}}{\frac{16}{3}} (x-3) = 6(x-3)$$

• pokračujeme l_3

$$l_3 = x^2 - \frac{31}{3} - 6(x-3) = x^2 - 6x + \frac{23}{3}$$

ortogonálna projekcia $\min x$

- hľadáme polynóm $p(x) = ax^2 + bx + c$, aby bol naj. sprbl. funkcií $\min x$ na intervale $[1, 5]$

$$1.) \int_1^5 (\min x - (ax^2 + bx + c)) dx = 0$$

$$2.) \int_1^5 (\min x - (ax^2 + bx + c)) x dx = 0$$

$$3.) \int_1^5 (\min x - (ax^2 + bx + c)) x^2 dx = 0$$

$$1.) \int_1^5 \min x dx = a \int_1^5 x^2 dx + b \int_1^5 x dx + c \int_1^5 1 dx = -\cos 5 + \cos 1 \approx 0,256$$

$$2.) \int_1^5 x \min x dx = a \int_1^5 x^3 dx + b \int_1^5 x^2 dx + c \int_1^5 x dx \approx -4,88$$

$$3.) \int_1^5 x^2 \min x dx = a \int_1^5 x^4 dx + b \int_1^5 x^3 dx + c \int_1^5 x^2 dx \approx -20,7$$

$$0,256 = 41,33a + 12b + 4c$$

$$-4,88 = 156a + 41,33b + 12c$$

$$-20,7 = 624,8a + 156b + 41,33c$$

$$a \approx -0,11$$

$$b \approx 0,87$$

$$c \approx -0,63$$

$$p(x) \approx -0,11x^2 + 0,87x - 0,63$$

norma rozdielu

$$(\min x - p(x))^2$$

$$\int_1^5 (\min x - p(x))^2 dx \approx 0,25$$

$$\|\min x - p(x)\| \approx 0,5$$